Lecture 13 (10/14)

Omitted variables, Simpson's paradox, and some proofs

Last time:

- RCT vs. Cohort study.
- We can control for additional variables in a multiple predictor regression model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$$

- For the test scores example, we related the single predictor regression coefficient β'_{hs} to the multiple predictor regression coefficients and the regression coefficient between predictors

$$\underbrace{\beta'_{\text{hs}}}_{11.8} = \underbrace{\beta_{\text{hs}}}_{5.95} + \underbrace{\beta_{\text{iq}}}_{0.56} \underbrace{\beta_{\text{hs,iq}}}_{10.3},$$

- To compare predictors with different units, compare $\hat{\beta}_k \sigma_{X_k}$.

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Today:

1) General formulas: Omitted variable bias and ρ^2
2) More examples
3) Assignment 2

Theory

Theorem

Suppose $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$, where ϵ is uncorrelated with X_1 and X_2, and let $\rho^2 = 1 - \sigma_\epsilon^2 / \sigma_Y^2$.

(1) *Omitted variable bias: if X_2 is left out, the slope of the regression of Y on X_1 is*

$$\boxed{\beta_1' = \beta_1 + \beta_2 \beta_{1,2}}, \qquad \beta_{1,2} = \frac{cov(X_1, X_2)}{var(X_1)}.$$

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(2) *Adding a predictor never decreases ρ^2.*

The same argument with sample covariances shows R^2 never decreases. Both results extend to K predictors.

Proof of (1)

For simplicity, assume X_1, X_2 have mean zero and all intercepts are zero. Let $Y = \beta_1 X_1 + \beta_2 X_2 + \epsilon$ and let β_1' be the slope of the regression of Y on X_1:

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Using bilinearity of covariance and $\operatorname{cov}(X_1, \epsilon) = 0$,

$$\begin{aligned} \operatorname{cov}(X_1, Y) &= \beta_1 \operatorname{cov}(X_1, X_1) + \beta_2 \operatorname{cov}(X_1, X_2) + \operatorname{cov}(X_1, \epsilon) \\ &= \beta_1 \operatorname{var}(X_1) + \beta_2 \operatorname{cov}(X_1, X_2). \end{aligned}$$

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Dividing by $\operatorname{var}(X_1)$,

$$\beta_1' = \beta_1 + \beta_2 \beta_{1,2}, \qquad \beta_{1,2} = \frac{\operatorname{cov}(X_1, X_2)}{\operatorname{var}(X_1)}.$$

Proof of (2)

Again assume mean-zero predictors and zero intercepts. Regress X_2 on X_1: write $X_2 = \beta_{1,2} X_1 + \epsilon'$, where $\text{cov}(X_1, \epsilon') = \text{cov}(X_1, X_2) - \beta_{1,2}\text{var}(X_1) = 0$.

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$$Y = \underbrace{(\beta_1 + \beta_2 \beta_{1,2})}_{\beta_1'} X_1 + (\beta_2 \epsilon' + \epsilon).$$

Since $\text{cov}(X_1, \beta_2 \epsilon' + \epsilon) = 0$, this is the single-predictor regression of Y on X_1.

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$$\text{var}(\beta_2 \epsilon' + \epsilon) = \beta_2^2 \,\text{var}(\epsilon') + \sigma_\epsilon^2 + 2\beta_2 \underbrace{\text{cov}(\epsilon', \epsilon)}_{0} \;\geq\; \sigma_\epsilon^2,$$

where $\text{cov}(\epsilon', \epsilon) = 0$ because ϵ' is a linear function of X_1, X_2.

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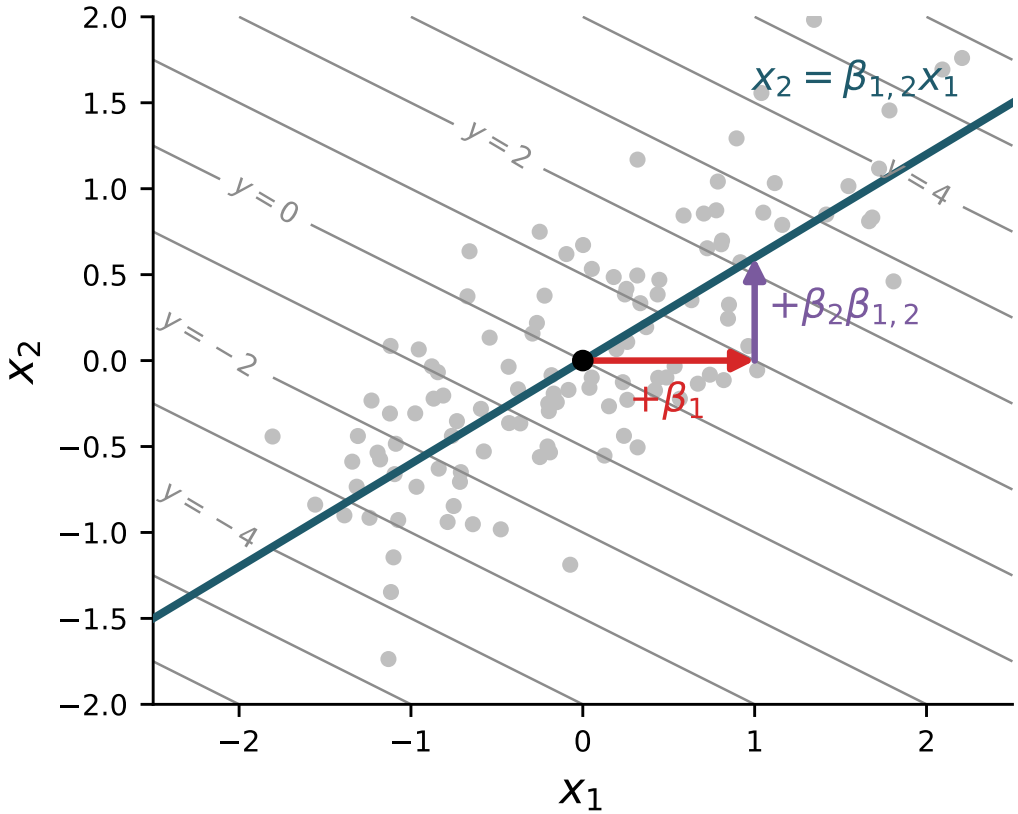
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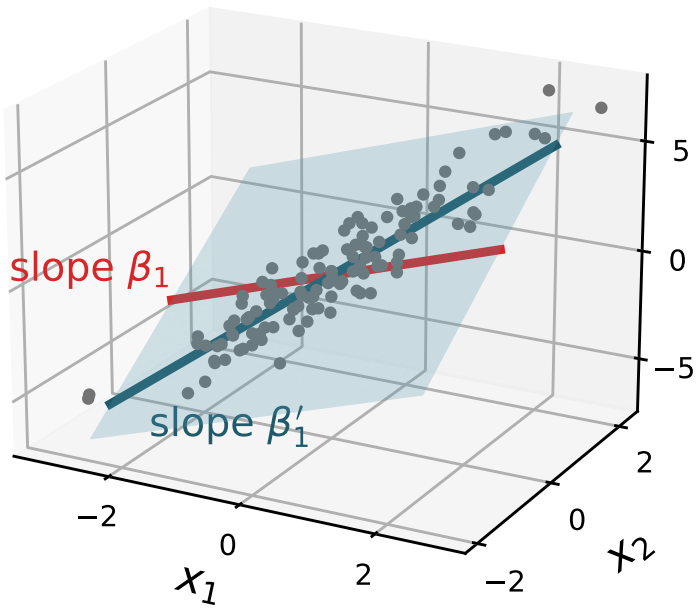
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where $\text{cov}(\epsilon', \epsilon) = 0$ because ϵ' is a linear function of X_1, X_2. Therefore

$$\rho'^2 = 1 - \frac{\text{var}(\beta_2 \epsilon' + \epsilon)}{\sigma_Y^2} \;\leq\; 1 - \frac{\sigma_\epsilon^2}{\sigma_Y^2} = \rho^2.$$

Visualizing OVB with 2 continuous predictors





Example

Data for the 50 US states in 2009 (from `statsmodels`; DC excluded):

- Y (`violent`): violent crimes per 100,000 people;
- X_{pov} (`poverty`): % of people below the poverty line;
- X_{single} (`single`): % of family households headed by a single parent.

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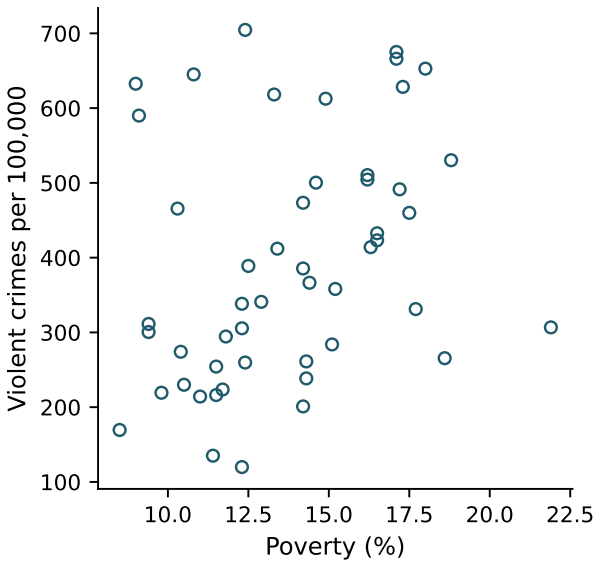
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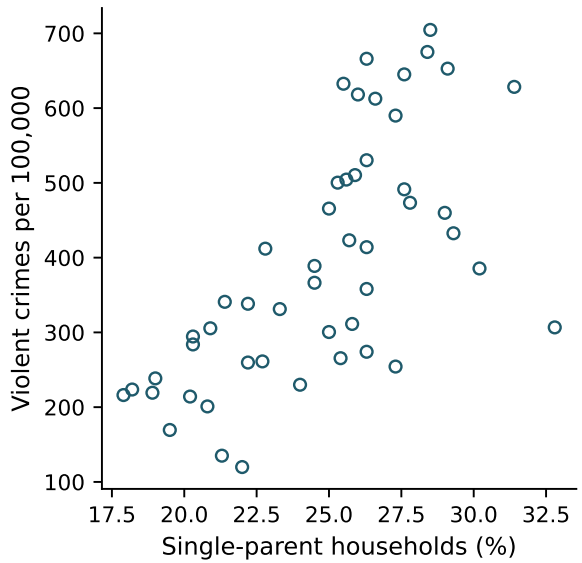
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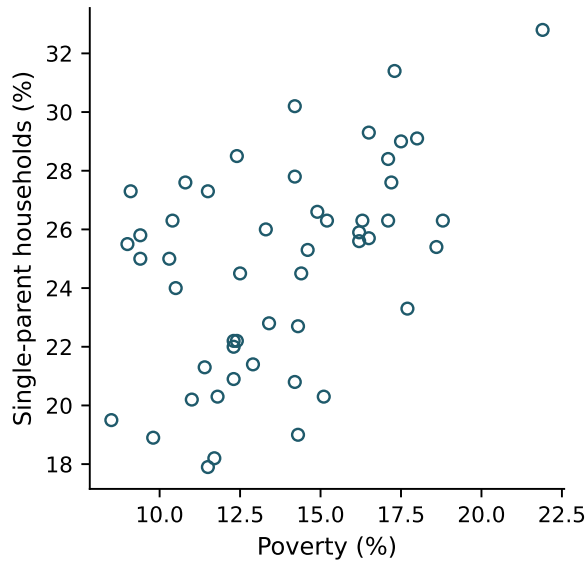
We fit three regression models:

$$
\begin{aligned}
Y &= \beta_0' + \beta_{\text{pov}}' X_{\text{pov}} + \epsilon, \\
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Poverty, single-parent households, and violent crime







Each point is one state (2009; DC excluded).

Dep. Variable:		violent	R-squared:			0.093
	coef	std err	t	P>|t|	[0.025	0.975]
const	172.9711	101.400	1.706	0.095	-30.906	376.849
poverty	15.9665	7.193	2.220	0.031	1.503	30.430

Dep. Variable:		violent	R-squared:			0.436
	coef	std err	t	P>|t|	[0.025	0.975]
const	-348.9463	122.900	-2.839	0.007	-596.054	-101.838
single	29.9911	4.920	6.095	0.000	20.098	39.884

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Question

Interpret the regression coefficients, and find $\beta_{\text{pov,single}}$, the slope when `single` is regressed on `poverty`. Can you deduce the correlation coefficient between the two predictors from this output?

Who is the better hitter?

Batting average = hits / at bats, for two New York Yankees:

	1995	1996	both years
Derek Jeter	$12/48 = .250$	$183/582 = .314$	
David Justice	$104/411 = .253$	$45/140 = .321$	

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Question

Explain this in terms of omitted variable bias. What is the omitted variable?

Conditional means and the omitted year

Let $Y = 1$ for a hit and $Y = 0$ otherwise, $X_{\text{jeter}} = 1$ for Jeter (0 for Justice), and $X_{\text{year}} = 1$ for 1996 (0 for 1995). We consider a model

$$E[Y \mid X_{\text{jeter}}, X_{\text{year}}] = \beta_0 + \beta_{\text{jeter}} X_{\text{jeter}} + \beta_{\text{year}} X_{\text{year}}$$

where the table gives

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\beta_0 &= E[Y \mid X_{\text{jeter}} = 0, X_{\text{year}} = 0] = 104/411 = .253, \\
\beta_0 + \beta_{\text{jeter}} &= E[Y \mid X_{\text{jeter}} = 1, X_{\text{year}} = 0] = 12/48 = .250, \\
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The table also gives us

$$\begin{aligned}
E[X_{\text{year}} \mid X_{\text{jeter}} = 1] &= 582/630 = .924, \\
E[X_{\text{year}} \mid X_{\text{jeter}} = 0] &= 140/551 = .254.
\end{aligned}$$

Other examples of Simpson's paradox

- **Graduate admissions, UC Berkeley 1973:** men were admitted at a higher rate overall (44% vs. 35%), but in most departments women were admitted at a similar or higher rate. Omitted variable: the department; women applied more often to departments that admitted few applicants.

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- **COVID-19, early 2020:** the case fatality rate was higher in Italy than in China overall, but lower in every age group. Omitted variable: age; Italy's cases were older.

Question

Suppose we regress Y_t on *two* previous values, Y_{t-1} and Y_{t-2}:

$$Y_t = \beta_0 + \beta_{\text{lag1}} Y_{t-1} + \beta_{\text{lag2}} Y_{t-2} + \epsilon.$$

What should β_{lag1} and β_{lag2} be? What if we regress Y_t on Y_{t-2} alone?

Solutions

- **Poverty and crime:** $\hat{\beta}_{\text{pov,single}} = 0.565$, $\hat{\beta}_{\text{single,pov}} = 0.425$, $\hat{\rho} = 0.49$.
- **Batting averages:** $\hat{\beta}'_{\text{jeter}} = -0.006 + 0.067 \times 0.670 = 0.039$.
- **Two lags** (AR(1) with coefficient ρ): $\beta_{\text{lag1}} = \rho$, $\beta_{\text{lag2}} = 0$; on Y_{t-2} alone, $\beta'_{\text{lag2}} = \rho^2$.